GATE 2027 CS/IT
MASTER INDEX & PYQ COVERAGE INVENTORY

All 13 Subject Books + General Aptitude • Chapter / Subtopic / Year Map • Repository Source Audit

Source Repository: *kl2400032185/STUDY-need* — cloned 23 Oct 2026 | **Papers Processed:** 18 CS papers (2016 Set-1/2, 2017 Set-1/2 scan, 2018, 2019, 2020 scan, 2021 Set-1/2, 2022, 2023, 2024 Set-1/2, 2025 Set-1/2, 2026 Set-1/2) | **Answer Keys:** 18/18 text-extractable | **Question Papers:** 16/18 text-extractable (2017,2020 image-scanned — visually verifiable) | **Total Questions in Scope:** ~1,170 (65 per paper × 18) | **GA per Paper:** 10 Qs (5×1M +5×2M =15 marks verified)

DELIVERABLE INVENTORY — PDF LIST

#	PDF File	Subject	Chapters	PYQs Solved (in book)	Pages (approx)
01	01_Programming_and_Data_Structures.pdf	Programming & Data Structures	3	7	~22
02	02_Algorithms.pdf	Algorithms	3	4	~20
03	03_Discrete_Mathematics.pdf	Discrete Mathematics	3	3	~18
04	04_Engineering_Mathematics_Linear_Algebra.pdf	Engineering Maths — Linear Algebra	2	3	~16
05	05_Engineering_Mathematics_Probability_and_Statistics.pdf	EM — Probability & Statistics	2	2	~15
06	06_Engineering_Mathematics_Calculus.pdf	EM — Calculus	2	2	~15
07	07_Digital_Logic.pdf	Digital Logic	2	2	~15
08	08_Computer_Organization_and_Architecture.pdf	COA	2	2	~15
09	09_Operating_Systems.pdf	Operating Systems	2	2	~15
10	10_Theory_of_Computation.pdf	Theory of Computation	2	2	~17
11	11_Computer_Networks.pdf	Computer Networks	2	2	~16
12	12_Databases_(DBMS).pdf	Databases (DBMS)	2	2	~16
13	13_Compiler_Design.pdf	Compiler Design	2	2	~16
14	14_General_Aptitude_GATE_2027.pdf	General Aptitude (Official Syllabus 4 Sections)	4	8	~28
	00_MASTER_INDEX_GATE_2027.pdf	This file — Master Index & Coverage Audit	—	—	~6
	PYQ_Coverage_Spreadsheet.csv	Spreadsheet: subject/topic/year/paper/marks/type/answer/solution	—	All 18 papers	—

File naming: Files are numbered 01–14 for study order priority (Programming → DM → Algorithms → GA first). All PDFs are A4, printable, with headers/footers, page numbers, clickable TOC, and methodology note. Open any PDF with any viewer (Adobe, Chrome, Foxit, mobile).

CHAPTER & SUBTOPIC MAP — ALL SUBJECTS

Book	Chapters & Subtopics (per GATE 2027 syllabus)
01 PDS	C Programming/Pointers/Arrays; Recursion/Complexity; Stacks/Queues/LL; Trees/BST/Heaps/Hashing/Graphs
02 Algorithms	Asymptotics & Recurrences; D&C; & Greedy; DP & Graph Algorithms
03 Discrete Maths	Logic; Counting & Recurrences; Graph Theory (Euler/Hamilton/Planar/Colouring)
04 Linear Algebra	Matrices/Rank/Determinant; Linear Systems & Eigen Theory
05 Probability & Stats	Conditional/Bayes; RVs & Distributions & Statistics
06 Calculus	Limits/Continuity/Maxima-Minima; Integration & Vector Calculus
07 Digital Logic	Boolean/K-map/Combinational; Sequential (FF/Counter/FSM)
08 COA	Instructions/ALU/Pipelining/Hazards; Cache & Virtual Memory
09 OS	Processes/Scheduling/Synchronization; Memory & Deadlocks

10 TOC	Regular Languages; CFL/TM/Decidability
11 CN	Data Link (Framing/Error/Flow); Network & Transport (IP/Routing/TCP)
12 DBMS	Relational/ER/Algebra/SQL; Normalization/Transactions/Concurrency
13 Compiler	Lexical & Parsing; Translation/CodeGen/Optimization
14 GA	Verbal; Quantitative; Analytical; Spatial (Official 2027 4-section syllabus)

PAPER COVERAGE — WHAT WAS PROCESSED VS WHAT IS MISSING

The repository supplied by the user contained **18 GATE CS papers (2016–2026)** with both question papers and answer keys. Papers before 2016 (1991–2015) are **not in the supplied files** and therefore not included in the current PYQ counts; they are listed as *Future Scope*. Pre-2016 papers can be added in a v1.1 update when files are provided — the existing PDFs explicitly flag this so you know the boundary.

Year/Paper	Q-Paper File	Answer Key	Format	Status	Notes
2016 Set-1	GATE_2016_CS_Set-1_Question_Paper.pdf	GATE_2016_CS_Set-1_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs; MCQ/NAT/Marks verified
2016 Set-2	GATE_2016_CS_Set-2_Question_Paper.pdf	GATE_2016_CS_Set-2_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs; fully extractable
2017 Set-1/2	GATE_2017_CS_Set-1/2_Question_Paper.pdf	GATE_2017_CS_Set-1/2_Answer_Key.pdf	Scanned Image	■ Partial	Answer keys fully processed; question text from scan needs visual figure check
2018	GATE_2018_CS_Question_Paper.pdf	GATE_2018_CS_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs
2019	GATE_2019_CS_Question_Paper.pdf	GATE_2019_CS_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs
2020	GATE_2020_CS_Question_Paper.pdf	GATE_2020_CS_Answer_Key.pdf	Scanned Image	■ Partial	As 2017 — answer keys OK
2021 Set-1/2	GATE_2021_CS_Set-1/2_Question_Paper.pdf	GATE_2021_CS_Set-1/2_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs each
2022	GATE_2022_CS_Question_Paper.pdf	GATE_2022_CS_Answer_Key.pdf	Text PDF	✓ Processed	66 Qs (bonus)
2023	GATE_2023_CS_Question_Paper.pdf	GATE_2023_CS_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs
2024 Set-1/2	GATE_2024_CS_Set-1/2_Question_Paper.pdf	GATE_2024_CS_Set-1/2_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs each
2025 Set-1/2	GATE_2025_CS_Set-1/2_Question_Paper.pdf	GATE_2025_CS_Set-1/2_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs each
2026 Set-1/2	GATE_2026_CS_Set-1/2_Question_Paper.pdf	GATE_2026_CS_Set-1/2_Answer_Key.pdf	Text PDF	✓ Processed	65 Qs each
1991–2015	— not in repo —	— not in repo —	—	Future Scope	Not supplied; available for v1.1 when files provided

Accuracy note: Q numbers, marks (1M/2M) and types (MCQ/MSQ/NAT) in every book come from the official answer-key 'Type/Marks' columns, not from guessing. MCQ/MSQ options and NAT ranges are reproduced from keys. Where question-statement extraction was lossy (scanned 2017/2020), the book uses the answer-key Q number + marks/type and paraphrases the stem from the official paper image, flagged 'verify figure' — never fabricated. Future v1.1 will add pre-2016 papers when supplied.

DOWNLOADS & USAGE

All PDFs are stored in **gate2027_pdfs/** in this repository (branch *arena/5a3f4d7d-youtube1*). Download each PDF from the GitHub file browser (click → Download) or via the raw links below. For best print, use A4, Colour or Grayscale both work (high contrast), and enable 'Fit to page'. The CSV spreadsheet can be opened in Excel / Google Sheets.

Recommended Study Priority (for Rank <100):

- **Phase 1 (Weeks 1-6):** PDS → Algorithms → Discrete Maths → General Aptitude (daily 1 GA mock).
- **Phase 2 (Weeks 7-12):** OS → DBMS → TOC → CN (alternate theory day + PYQ day).
- **Phase 3 (Weeks 13-18):** Linear Algebra → Probability/Stats → Calculus → Digital Logic → COA → Compiler.
- **Phase 4 (Weeks 19-24):** Full-syllabus timed mocks (3 hrs, 65 Qs) + Rapid Revision Sheets only.

GitHub raw download links (this branch):

Base: **https://github.com/kl2400032185/youtube1/blob/arena/5a3f4d7d-youtube1/gate2027_pdfs/<filename>** — or raw:
https://raw.githubusercontent.com/kl2400032185/youtube1/arena/5a3f4d7d-youtube1/gate2027_pdfs/<filename>

- **01_Programming_and_Data_Structures.pdf** — 46 KB | gate2027_pdfs/01_Programming_and_Data_Structures.pdf
- **02_Algorithms.pdf** — 40 KB | gate2027_pdfs/02_Algorithms.pdf
- **03_Discrete_Mathematics.pdf** — 39 KB | gate2027_pdfs/03_Discrete_Mathematics.pdf
- **04_Engineering_Mathematics_Linear_Algebra.pdf** — 32 KB | gate2027_pdfs/04_Engineering_Mathematics_Linear_Algebra.pdf
- **05_Engineering_Mathematics_Probability_and_Statistics.pdf** — 31 KB | gate2027_pdfs/05_Engineering_Mathematics_Probability_and_Statistics.pdf
- **06_Engineering_Mathematics_Calculus.pdf** — 31 KB | gate2027_pdfs/06_Engineering_Mathematics_Calculus.pdf
- **07_Digital_Logic.pdf** — 30 KB | gate2027_pdfs/07_Digital_Logic.pdf
- **08_Computer_Organization_and_Architecture.pdf** — 31 KB | gate2027_pdfs/08_Computer_Organization_and_Architecture.pdf
- **09_Operating_Systems.pdf** — 32 KB | gate2027_pdfs/09_Operating_Systems.pdf
- **10_Theory_of_Computation.pdf** — 35 KB | gate2027_pdfs/10_Theory_of_Computation.pdf
- **11_Computer_Networks.pdf** — 31 KB | gate2027_pdfs/11_Computer_Networks.pdf
- **12_Databases_(DBMS).pdf** — 33 KB | gate2027_pdfs/12_Databases_(DBMS).pdf
- **13_Compiler_Design.pdf** — 32 KB | gate2027_pdfs/13_Compiler_Design.pdf
- **14_General_Aptitude_GATE_2027.pdf** — 52 KB | gate2027_pdfs/14_General_Aptitude_GATE_2027.pdf

Good luck — solve honestly, revise ruthlessly. Rank <100 is built one PYQ at a time.